SE444: Introduction to Artificial Intelligence

Mock Final Exam — Answer Sheet

Fall 2025

Date: TBD, January 2026

Student Information

Student ID:		Student Name:	

Part I. Multiple Choice (6 pts, 24 min)

Instructions: Type **X** in the box for correct answer(s). Each question may have 1 or more correct answers.

Opt	Q1	Q2	Q3	Q4	Q5	Q6
A						
B						
C						
D						
E						

Part II. Probability & BN (11 pts, 33 min)

Question II.1: BN Construction & d-Separation (4 pts, 12 min)

Scenario: Cybersecurity Intrusion Detection with 5 variables.

(a) BN Structure [2 pts]:

(b) Is $I \perp S \mid F$? d-Separation analysis [1 pt]:

(c) Collider + Explaining Away [1 pt]:

Question II.2: Bayesian Inference (4 pts, 12 min)

Smart Home: $H \to M \to L$ with given CPTs.

(a) Compute $P(H = T \mid L = T)$ by enumeration [2 pts]:

(b) Variable Elimination with factor tables [2 pts]:

Question II.3: Decision Under Uncertainty (3 pts, 9 min)

Drone Delivery: Weather probabilities and utility matrix given.

(a) Expected Utility for each action [2 pts]:

EU(Fly Direct) =

EU(Take Detour) =

EU(Wait) =

(b) MEU-optimal action [1 pt]:

Part III. MDPs & Value Iteration (5 pts, 15 min)

Question III.1: MDP & Value Iteration (5 pts, 15 min)

Warehouse Robot: 2×3 grid, Goal=+10, Living=−1, $\gamma = 0.9$, stochastic transitions.

(a) MDP Components (S, A, P, R, γ) **[1 pt]:**

(b) Value Iteration — 2 iterations for S3 and S5 [2 pts]:

Iteration 1:

$V_1(S3) =$

$V_1(S5) =$

Iteration 2:

$V_2(S3) =$

$V_2(S5) =$

(c) Optimal policy and justification [2 pts]:

$\pi^*(S3) =$ $\pi^*(S5) =$

Justification:

Part IV. Planning & Game Theory (6 pts, 18 min)

Question IV.1: STRIPS Planning (3 pts, 9 min)

E-Commerce Fulfillment: Initial/Goal states given.

(a) STRIPS Load(pkg,truck,dock) [1 pt]:

Preconditions:

Add List:

Delete List:

(b) Forward Search Plan [2 pts]:

Step 1:

Step 2:

Step 3:

Step 4: (if needed)

Question IV.2: Nash Equilibrium (3 pts, 9 min)

Tech Startup Competition: 3×3 payoff matrix given.

(a) Dominated Strategies / IESDS [1 pt]:

s

(b) Pure-Strategy Nash Equilibria [2 pts]:
Mark best responses in matrix, then list NE:

A \ B	Premium	Budget	Freemium
Premium	(,)	(,)	(,)
Budget	(,)	(,)	(,)
Freemium	(,)	(,)	(,)

Nash Equilibrium/Equilibria:

Part V. Project Code (5 pts, 12 min)

Question V.1: Search Algorithm Analysis (3 pts, 6 min)

Given: A* search code snippet.

(a) g_score, f_score, h() meaning & $f(n) = g(n) + h(n)$ [1 pt]:

(b) Purpose of "if current in explored: continue" [1 pt]:

(c) Is Manhattan distance admissible for 4-connected grid? [1 pt]:

Question V.2: Bayesian Belief Updates (2 pts, 6 min)

Given: Belief update code with prior, sensor_reading, accuracy.

(a) Compute posterior (prior=0.3, detect=True, accuracy=0.9) [1 pt]:

(b) Effect when accuracy=0.5 [1 pt]:

FINAL EXAM TOTAL: _____ **/ 30**